Zhengjie Min

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Education

University of Michigan Expected May 2027
B.S.E. in Computer Science; Mathematics minor in progress *GPA: **3.98**; Major: **4.00***

Shanghai Jiao Tong University Expected August 2027
Bachelor of Engineering in Mechanical Engineering *GPA: **3.77***

Research Interests

Graph algorithms, parameterized algorithms (FPT), combinatorial optimization, matroids, and combinatorics; computational complexity and cryptography.

Research Projects

Connectivity Oracles under Edge Failures June 2026 – Present
Research discussions with Prof. Thatchaphol Saranurak, University of Michigan. Write-up available upon request.

Metric Minimum-Cost Perfect Matching January 2026 – Present
Research discussions with Prof. Seth Pettie, University of Michigan

- Investigated candidate obstructions to odd-blossom extraction involving odd cuts and all-pairs distances; wrote a technical report on a proposed approximation approach.
- Exploring a $(1+\varepsilon)$-approximation targeting near-$O(m/\varepsilon)$ time.

Selected Course Projects

Robust Pseudorandom Generation | *Advanced Cryptography* October 2026
Course instructor: Prof. Christopher Peikert, University of Michigan

- Analyzed the Barak and Halevi generator, relaxing its extraction premise to negligible statistical distance and bounding error across repeated refreshes.
- Constructed a seed-bit leakage example separating robustness from refresh-input privacy after state exposure; formulated a prediction-based privacy goal.

Randomized k-Path Algorithms | *Parameterized Algorithms* December 2025
Course instructor: Prof. Euiwoong Lee, University of Michigan

- Proved a high-probability polynomial-time guarantee for a simple random-walk algorithm for k-Path when the average degree is at least $6k^3$.

Selected Theory Coursework

In progress, Fall 2026: Computational Complexity (CSE 574); Advanced Cryptography (CSE 575); Combinatorial Optimization in Graph Algorithms (CSE 598); Combinatorics and Graph Theory (MATH 565).

Completed: Introduction to Algorithms (A); Parameterized Algorithms (A); Zero-Knowledge Proofs (A); Foundations of Computer Science (A+); Introduction to Probability (A+); Introduction to Cryptography (A+).

Honors & Awards

Jackson and Muriel Lum Endowed Scholarship July 2025 & July 2026
University of Michigan College of Engineering

ICPC North America East Central Regional Contest | Rank 31 November 2025

Selected Software Projects

The Ranking Hat | *C++ search engine, University of Michigan* January – May 2026

- Implemented query compilation, tuned ranking, and added automated code-quality checks for a six-person C++ search engine; the team indexed 480 GB with 0.7-second average query response time.

Joint Teapot | *Python automation, SJTU Global College* February 2025 – Present

- Developed a workflow integrating Gitea, Canvas, and Mattermost REST APIs to automate programming-assignment submission and grading.

Technical Skills

Languages: C++, Python, C, Rust, Lean 4, Shell, Elm, MATLAB.
Tools: LaTeX, Typst, Git, Linux, CMake, GitHub CI.